

ELEC 221 Lecture 06

CT Fourier series coefficients and properties

Tuesday 29 September 2026

Announcements

- Quiz 3 today
- Practice Assignment 2 available
- Tutorial Assignment 2 is **next week**.

Last time

Complex exponential signals are eigenfunctions of LTI systems:

$$x(t) = e^{st} \quad y(t) = H(s) e^{st}$$

$H(s)$ is the **system function**.

$$H(s) = \int_{-\infty}^{\infty} e^{-s\tau} h(\tau) d\tau$$

Last time

When we restricted to complex values only, i.e.,

$$s = j\omega \Rightarrow x(t) = e^{j\omega t} = \cos \omega t + j \sin \omega t$$

the system function is called the **frequency response**,

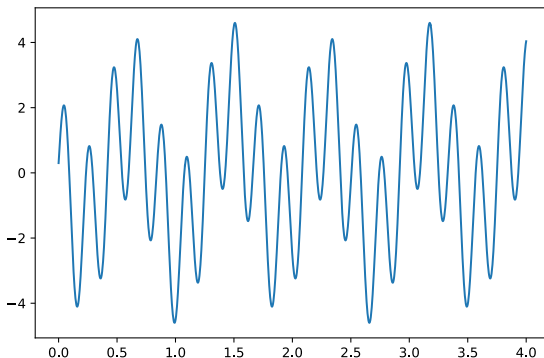
$$H(j\omega) = \int_{-\infty}^{\infty} e^{-j\omega\tau} h(\tau) d\tau$$

If a superposition of such signals is input into an LTI system, each signal is rescaled by the frequency response:

$$x(t) = \sum_k c_k e^{j\omega_k t} \rightarrow y(t) = \sum_k c_k H(j\omega_k) e^{j\omega_k t}$$

Last time

What if we have a signal like this:



Can it be expressed in terms of complex exponential signals?

Today

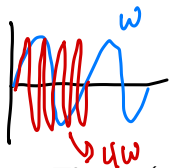
Learning outcomes:

- Express a periodic CT signal as a Fourier series and compute its Fourier series coefficients
- State the Dirichlet conditions and identify whether a signal can be expressed as a Fourier series
- Describe the Gibbs phenomenon
- Describe how Fourier coefficients are affected by signal transformations

The Fourier series

$$x(t) = e^{j\omega t} \rightarrow x_k(t) = e^{j\omega k t} \quad \uparrow \text{integer } k$$

Given $x(t)$ with period T (and some additional conditions) we can write it as a **Fourier series**.



$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega t}$$

The c_k (complex numbers) are its **Fourier coefficients**.

The Fourier series

Usually dealing with $x(t)$ that is always *real*.

$$x(t) = \sum_{k=-\infty}^{\infty} C_k e^{j k \omega t}$$

$$x^*(t) = \sum_{k=-\infty}^{\infty} C_k^* e^{-j k \omega t} = \sum_{k=-\infty}^{\infty} C_k e^{j k \omega t}$$

$k = -1 \Rightarrow C_{-1}^*$
 $k = 1 \Rightarrow C_k$

This means

$$C_k^* = C_{-k}$$

We can leverage this to express $x(t)$ in a different way.

The Fourier series

Apply Euler's relation:

$$\begin{aligned}x(t) &= \sum_{k=-\infty}^{\infty} c_k e^{j\omega_k t} \\&= \sum_{k=-\infty}^{\infty} c_k (\cos(\omega_k t) + j \sin(\omega_k t)) \\&= \sum_{k=-\infty}^{\infty} c_k \cos(\omega_k t) + j \sum_{k=-\infty}^{\infty} c_k \sin(\omega_k t)\end{aligned}$$

With a bit of work, you can derive

$$x(t) = a_0 + \sum_{k=1}^{\infty} (a_k \cos(\omega_k t) + b_k \sin(\omega_k t))$$

where a_0 , a_k , b_k are expressed in terms of the c_k .

Note: sometimes a_0 is written $a_0/2$ for convenience

We derive the relationship between coefficients in the two expressions for Fourier series below:

$$x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega t} = a_0 + \sum_{k=1}^{\infty} (a_k \cos(k\omega t) + b_k \sin(k\omega t))$$

Recall that $C_k^* = C_{-k}$.

Start by applying Euler's identity to $e^{jk\omega t}$:

$$x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega t} = \sum_{k=-\infty}^{\infty} C_k (\cos(k\omega t) + j \sin(k\omega t))$$

First, note that for $k=0$, $\cos(k\omega t)=1$, $\sin(k\omega t)=0$ so only the constant survives, i.e.,

$$a_0 = C_0$$

Next, let's look just at the cos terms:

$$\sum_{k=-\infty}^{\infty} C_k \cos(k\omega t) = \dots + C_{-k} \cos(-k\omega t) + \dots + C_{-1} \cos(-\omega t) + C_1 \cos(\omega t) + \dots + C_k \cos(k\omega t) + \dots$$

Since $\cos(-\theta) = \cos \theta$, we can write this as

$$= (C_1 + C_{-1}) \cos(\omega t) + \dots + (C_k + C_{-k}) \cos(k\omega t) + \dots$$

$$= \sum_{k=1}^{\infty} \underbrace{(C_k + C_{-k})}_{C_k + C_k^*} \cos(k\omega t)$$

$$C_k + C_{-k} = C_k + C_k^* = 2 \cdot \text{Re}(C_k)$$

Similarly, we can do this for $\sin(k\omega t) = -\sin(-k\omega t)$:

$$\sum_{k=-\infty}^{\infty} C_k \cdot j \sin(k\omega t) = \sum_{k=1}^{\infty} \underbrace{(C_k - C_{-k})}_{C_k - C_k^*} j \sin(k\omega t)$$

$$= \sum_{k=1}^{\infty} 2j \cdot \text{Im}(C_k) \cdot j \sin(k\omega t)$$

Thus, $a_0 = C_0$ $a_k = 2 \text{Re}(C_k)$ $b_k = -2 \text{Im}(C_k)$

We can also go "backwards". Start with the $\sin/\cos$ then apply Euler's relation:

$$\begin{aligned} a_0 + \sum_{k=1}^{\infty} (a_k \cos(k\omega t) + b_k \sin(k\omega t)) \\ = a_0 + \sum_{k=1}^{\infty} \left(a_k \left(\frac{e^{jk\omega t} + e^{-jk\omega t}}{2} \right) + b_k \left(\frac{e^{jk\omega t} - e^{-jk\omega t}}{2j} \right) \right) \\ = a_0 + \sum_{k=1}^{\infty} \left(\underbrace{\left(\frac{a_k}{2} + \frac{b_k}{2j} \right)}_{C_k} e^{jk\omega t} + \underbrace{\left(\frac{a_k}{2} - \frac{b_k}{2j} \right)}_{C_{-k} = C_k^*} e^{-jk\omega t} \right) \end{aligned}$$

Thus, $C_k = \frac{a_k}{2} + \frac{b_k}{2j}$

$\Rightarrow \frac{a_k}{2}$ is real part of C_k ,

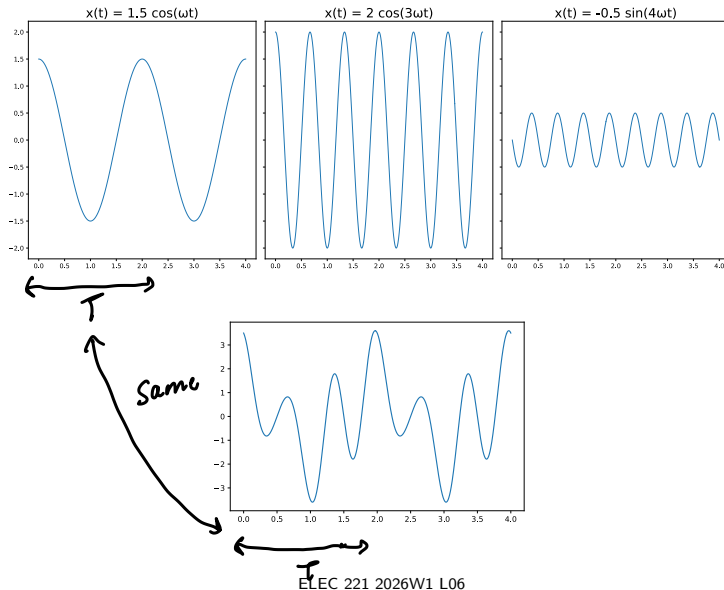
so $a_k = 2 \cdot \text{Re}(C_k)$

$-\frac{b_k}{2}$ is the imaginary part,

so $b_k = -2 \cdot \text{Im}(C_k)$

as derived on previous page.

The Fourier series



Evaluating Fourier coefficients

Given a signal $x(t)$, let's compute c_k .

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega t}$$

Handwritten red circle around c_k and red "???" below it.

The $e^{jk\omega t}$ are **basis functions** and have **orthogonality** relations.

Let $\phi_k(t) = e^{jk\omega t}$. Let's integrate over a period:

$$\int_0^T \phi_k(t) \phi_m^*(t) dt$$

where $*$ indicates the *complex conjugate*.

Evaluating Fourier coefficients

Exercise: evaluate the integral

$$\frac{1}{T} \int_0^T \phi_k(t) \phi_m^*(t) dt = \frac{1}{T} \int_0^T e^{jk\omega t} e^{-jm\omega t} dt$$

Case 1: $k = m$

$$\frac{1}{T} \int_0^T e^{j(k-m)\omega t} dt = \frac{1}{T} \int_0^T dt = 1$$

Case 2: $k \neq m$

$$\begin{aligned} \frac{1}{T} \int_0^T e^{j(k-m)\omega t} dt &= \frac{1}{T} \int_0^T \left[\cos((k-m)\omega t) + \frac{1}{T} j \int_0^T \sin((k-m)\omega t) dt \right] dt \\ &= 0 \end{aligned}$$

Evaluating Fourier coefficients

Exercise: express $x(t)$ as a Fourier series and evaluate the integral

$$\begin{aligned}
 & \frac{1}{T} \int_0^T \phi_m^*(t) x(t) dt \\
 & x(t) = \sum_{k=-\infty}^{\infty} C_k e^{jk\omega t} \\
 & \frac{1}{T} \int_0^T e^{-jm\omega t} x(t) dt = \frac{1}{T} \int_0^T \sum_{k=-\infty}^{\infty} C_k e^{-jm\omega t} e^{jk\omega t} dt \\
 & = \sum_{k=-\infty}^{\infty} \frac{1}{T} \int_0^T C_k e^{-jm\omega t} e^{jk\omega t} dt \\
 & = C_m \quad \text{($k=m$ survives)}
 \end{aligned}$$

Evaluating Fourier coefficients

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega t}$$

synthesis

$$\rightarrow c_k = \frac{1}{T} \int_T e^{-jk\omega t} x(t) dt$$

analysis

Fourier coefficients tell us *how much* each harmonic contributes.

Note that c_0 is a constant offset:

$$c_0 = \frac{1}{T} \int_T x(t) dt$$

(Similar techniques can be used to derive a_k and b_k for the sin and cos representation. Try it!)

Recap: key expressions

Fourier synthesis equation:

$$x(t) = \sum_{k=-\infty}^{\infty} c_k e^{jk\omega t}$$

Fourier analysis equation:

$$c_k = \frac{1}{T} \int_T e^{-jk\omega t} x(t) dt$$

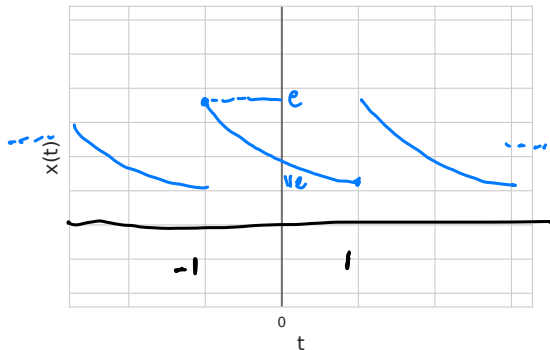
Exercise

What is the Fourier series of

$$x(t) = e^{-t}, \quad -1 \leq t < 1$$

Start with a plot, and determine T and ω .

$$T = 2$$
$$\omega = \pi$$



Exercise

$$\begin{aligned}
 C_k &= \frac{1}{T} \int_T e^{-j\omega_k t} x(t) dt \\
 &= \frac{1}{2} \int_{-1}^1 e^{-j\pi k t} e^{-t} dt \\
 &= \frac{1}{2} \int_{-1}^1 e^{-t(1+j\pi k)} dt \\
 &= \frac{1}{2} \frac{-1}{1+j\pi k} e^{-t(1+j\pi k)} \Big|_{-1}^1 \\
 &= \text{integrate...} \\
 &= \frac{1}{2(1+j\pi k)} \left(e^{1+j\pi k} - e^{-(1+j\pi k)} \right) \\
 &= \frac{(-1)^k}{2(1+j\pi k)} [e - e^{-1}]
 \end{aligned}$$

Handwritten notes: A blue circle around ω_k with an arrow pointing to π . A blue arrow points from the integral limits to the text "integrate over a period -1 to 1".

Dirichlet conditions

Can we *always* express a signal as a Fourier series?

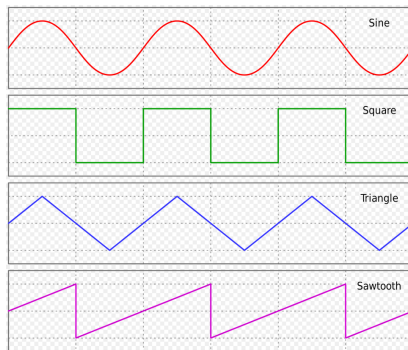


Image credit: *Sine, square, triangle, and sawtooth waveforms* (author: Omegatron)

https://en.wikipedia.org/wiki/Triangle_wave#/media/File:Waveforms.svg (CC BY-SA 3.0)

Dirichlet conditions

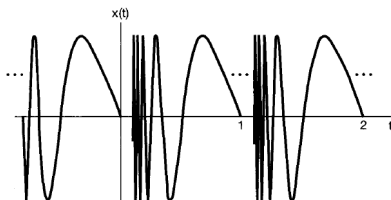
Given a **periodic function**, if over one period,

- ❶ is single-valued
- ❷ is absolutely integrable
- ❸ has a finite number of maxima and minima
- ❹ has a finite number of discontinuities

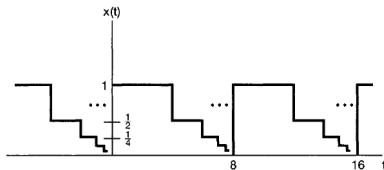
then the Fourier series converges to

- $x(t)$ where it is continuous
- half the value of the jump where it is discontinuous

Dirichlet conditions



(b)



(c)

Figure 3.8 Signals that violate the Dirichlet conditions: (a) the signal $x(t) = 1/t$ for $0 < t \leq 1$, a periodic signal with period 1 (this signal violates the first Dirichlet condition); (b) the periodic signal of eq. (3.57), which violates the second Dirichlet condition; (c) a signal periodic with period 8 that violates the third Dirichlet condition [for $0 \leq t < 8$, the value of $x(t)$ decreases by a factor of 2 whenever the distance from t to 8 decreases by a factor of 2; that is, $x(t) = 1$, $0 \leq t < 4$, $x(t) = 1/2$, $4 \leq t < 6$, $x(t) = 1/4$, $6 \leq t < 7$, $x(t) = 1/8$, $7 \leq t < 7.5$, etc.].

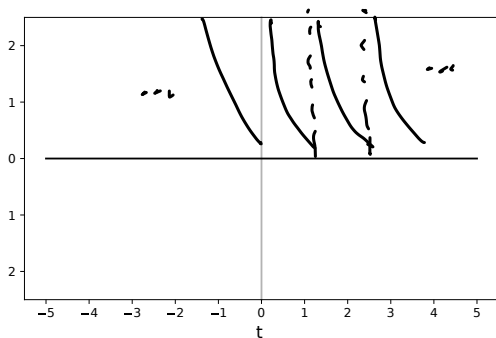
Dirichlet conditions

Exercise: can the periodic replication of the signal

$$x(t) = \frac{1}{t}, \quad 0 < t \leq 1$$

not absolutely integrable!

be expressed as a Fourier series?



For next time

Content:

- Discrete-time Fourier series coefficients and properties

Action items:

- ① Work on practice assignments 1 and 2

Recommended reading:

- From today's class: Oppenheim 3.0-3.5
- Suggested problems: 3.4, 3.5, 3.8, 3.13, 3.17, 3.22a,c, 3.23-3.26
- For next class: Oppenheim 3.6-3.7